

Name: _____

Class: _____

ACTIVITY SHEET

1.3 Smoke detectors

Read the article, and then answer the questions that follow.

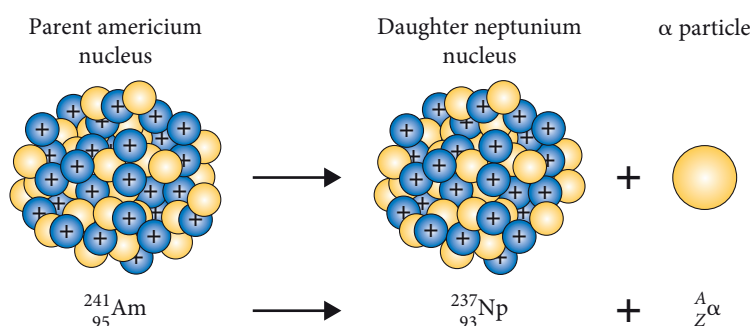
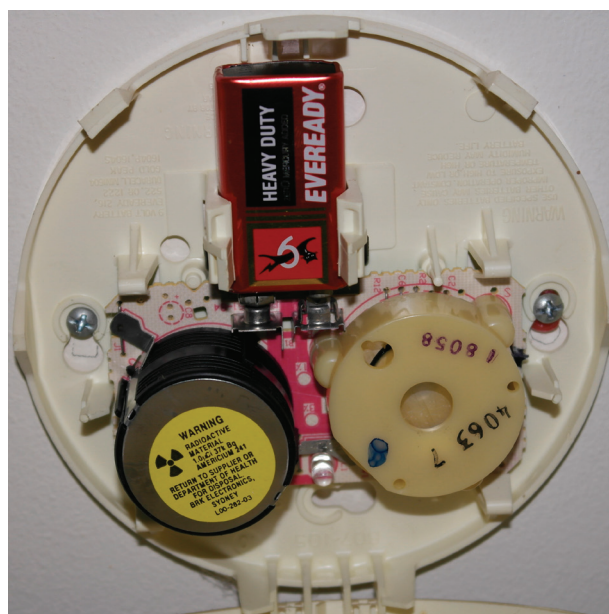
The photograph shows the interior of a smoke alarm or detector. It consists of an air-filled ionisation chamber (lower right) and a radioactive source of americium-241 (lower left). Americium (Am) is not a naturally occurring element; rather, it is a by-product of reactions occurring inside nuclear reactors.

Americium-241 is an unstable radioactive isotope and its nucleus will spontaneously eject alpha particles.

The alpha particles emitted by the americium isotope bombard the air inside the ionisation chamber, causing the air molecules to become electrically charged. This allows a small electric current to flow in the electrical field surrounding the chamber.

If smoke particles pass through the chamber, they also become ionised by alpha particles but because their mass is greater than that of air molecules, their acceleration in the electrical field is slower. This reduces the ionisation current, triggering a piezoelectric sounder, which emits a high-pitched squeal of about 85 decibels. (Sounds of about 75 decibels are needed to wake a sound sleeper.)

As the half-life* of the americium isotope is about 433 years, it will emit alpha particles well beyond the 10-year lifespan of the detector. When the americium atoms decay, they form an isotope of the element neptunium, which is much more stable with a half-life of 2.5 million years.



The ionising radiation of emitted alpha particles travels only a few centimetres in air before they are neutralised, so they are unlikely to come in contact with human body cells.

* Half-life is the amount of time needed for half of the nuclei in a radioactive sample to decay.

Questions

- 1 Describe the radioactive decay that occurs in a smoke detector.

- 2 What does the sign on the lower left chamber of the smoke detector in the photograph mean?

- 3 How many protons and neutrons does $^{241}_{95}\text{Am}$ have? What about $^{237}_{93}\text{Np}$?

- 4 What nuclear particles must an alpha particle consist of? How do you know?

- 5 What do alpha particles do to air and smoke molecules when they encounter them?

- 6 Why is a battery required in a smoke detector?

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